Day 13: Cut each part, multiply both numbers

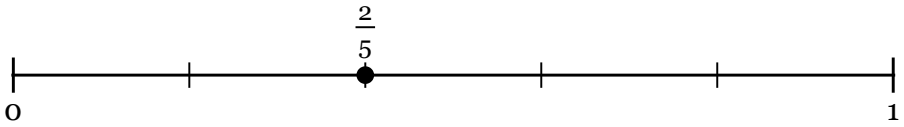
Name _____ Date _____

I can say what cutting each part does to both numbers, and write it as multiplying.

Warm-up.

1. Write the first six multiples of 7. _____
2. Find all the factor pairs of 18. _____

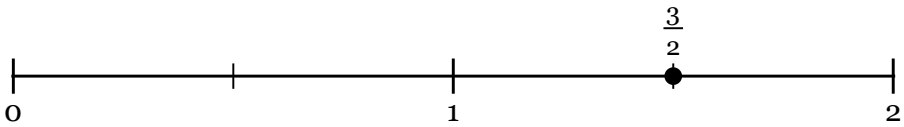
Example 1. The line from 0 to 1 is marked in fifths, with a point at $\frac{2}{5}$. Cut each fifth into 2 equal parts. What is the new name?



Each fifth was cut into ____, so I multiply both numbers by ____.

2/5 = (____ × 2)/(____ × 5) = ____/____

Example 2. The line from 0 to 2 is marked in halves, with a point at $\frac{3}{2}$. Cut each half into 3 equal parts. What is the new name?



Each half was cut into ____. 3/2 = (____ × 3)/(____ × 2) = ____/____

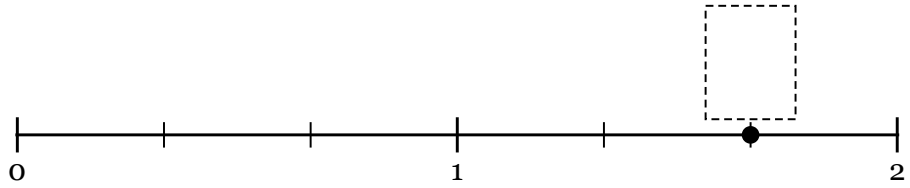
On your own. Mark the cuts on each line. Write the old name above the point and the new name below. Then write the rule.

3. Two lines from 0 to 1 are marked in halves, with a point at $\frac{1}{2}$. Cut each half into 3 on the first line. Cut each half into 5 on the second. Write each rule.



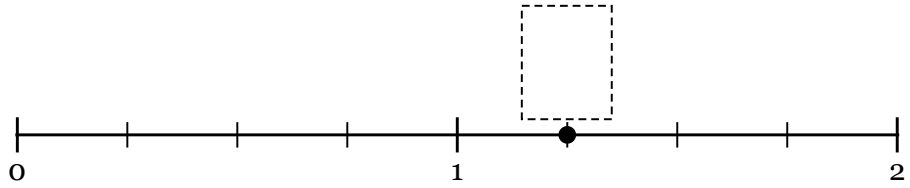
Rule: _____ Rule: _____

4. The line from 0 to 2 is marked in thirds, with a point at $\frac{5}{3}$. Cut each third into 2 equal parts. Write the rule.



Rule: _____

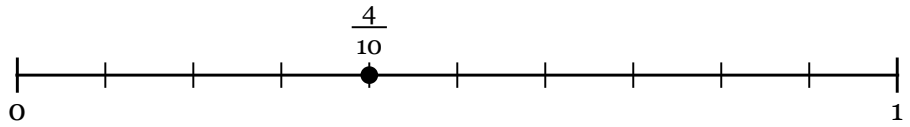
5. The line from 0 to 2 is marked in fourths, with a point at $\frac{5}{4}$. Cut each fourth into 2 equal parts. Write the rule.



Rule: _____

Fix it. The answer is wrong. Say what went wrong. Then do it right.

6. Ria cut each fifth into 2 equal parts. She wrote $\frac{2}{5} = \frac{2}{10}$. Her line shows the point at $\frac{4}{10}$.



What went wrong: _____

Right rule: _____

7. Sam says $\frac{4}{10}$ is twice as much as $\frac{2}{5}$, because both numbers were multiplied by 2.

What went wrong: _____

Challenge. A point is at $\frac{5}{3}$. Each third is cut into equal parts. The new name is __/12. Into how many parts was each third cut? Finish the name. _____